AC Voltage Controllers

AC to ac Converters

5.1 INTRODUCTION

An ac voltage controller is a converter that controls the voltage, current, and average power delivered to an AC load from an AC source. Electronic switches connect and disconnect the source and the load at regular intervals. In a switching scheme called phase control, switching takes place during every cycle of the source, in effect removing some of the source waveform before it reaches the load. Another type of control is integral-cycle control, whereby the source is connected and disconnected for several cycles at a time.

The phase-controlled AC voltage controller has several practical uses including light-dimmer circuits and speed control of induction motors. The input voltage source is AC, and the output is AC (although not sinusoidal), so the circuit is classified as an AC-AC converter.

5.2 The Single-Phase AC Voltage Controller

Basic Operation

A basic single-phase voltage controller is shown in Fig. 5-1*a*.

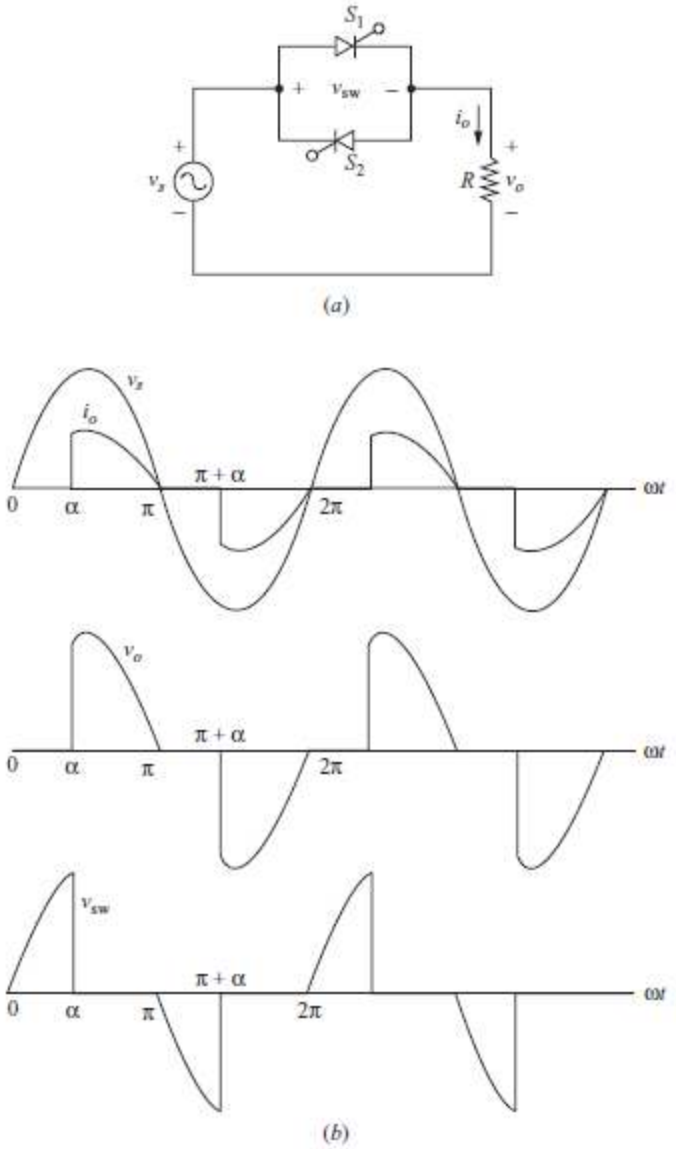


Figure 5-1 (*a*) Single-phase ac voltage controller with a resistive load; (*b*) Waveforms.

The electronic switches are shown as parallel thyristors (SCRs). This SCR arrangement makes it possible to have current in either direction in the load. This SCR connection is called antiparallel or inverse parallel because the SCRs carry current in opposite directions. A triac is equivalent to the antiparallel SCRs. Other controlled switching devices can be used instead of SCRs.

The principle of operation for a single-phase AC voltage controller using phase control is quite similar to that of the controlled half-wave rectifier of Sec. 3.9.

Here, load current contains both positive and negative half-cycles. An analysis identical to that done for the controlled half-wave rectifier can be

done on a half cycle for the voltage controller. Then, by symmetry, the result can be extrapolated to describe the operation for the entire period.

Some basic observations about the circuit of Fig. 5-1a are as follows:
1. The SCRs cannot conduct simultaneously.
2. The load voltage is the same as the source voltage when either SCR is on. The load voltage is zero when both SCRs are off.
3. The switch voltage v_{sw} is zero when either SCR is on and is equal to the source voltage when neither is on.
4. The average current in the source and load is zero if the SCRs are on for equal time intervals. The average current in each SCR is not zero because of the unidirectional SCR current.
5. The rms current in each SCR is times the rms load current if the SCRs are on for equal time intervals. (Refer to Chap. 2.)

For the circuit of Fig. 5-1a, S_1 conducts if a gate signal is applied during the positive half-cycle of the source. Just as in the case of the SCR in the controlled half-wave rectifier, S_1 conducts until the current in it reaches zero. Where this circuit differs from the controlled half-wave rectifier is when the source is in its negative half-cycle. Agate signal is applied to S_2during the negative half-cycle of the
source, providing a path for negative load current. If the gate signal for S_2 is a half period later than that of S_1, analysis for the negative half-cycle is identical to that for the positive half, except for algebraic sign for the voltage and current.

Single-Phase Controller with a Resistive Load

Figure 5-1b shows the voltage waveforms for a single-phase phase controlled voltage controller with a resistive load. These are the types of waveforms that exist in a common incandescent light-dimmer circuit. Let the source voltage be:

$$v_s(\omega t) = V_m \sin \omega t \qquad (5\text{-}1)$$

Output voltage is:

$$v_o(\omega t) = \begin{cases} V_m \sin \omega t & \text{for } \alpha < \omega t < \pi \text{ and } \alpha + \pi < \omega t < 2\pi \\ 0 & \text{otherwise} \end{cases} \qquad (5\text{-}2)$$

The rms load voltage is determined by taking advantage of positive and negative symmetry of the voltage waveform, necessitating evaluation of only a half period of the waveform:

$$V_{o,\text{rms}} = \sqrt{\frac{1}{\pi}\int_{\alpha}^{\pi}[V_m \sin(\omega t)]^2 d(\omega t)} = \frac{V_m}{\sqrt{2}}\sqrt{1 - \frac{\alpha}{\pi} + \frac{\sin(2\alpha)}{2\pi}} \quad (5\text{-}3)$$

Note that for $\alpha = 0$, the load voltage is a sinusoid that has the same rms value as the source. Normalized rms load voltage is plotted as a function of α in Fig. 5-2.

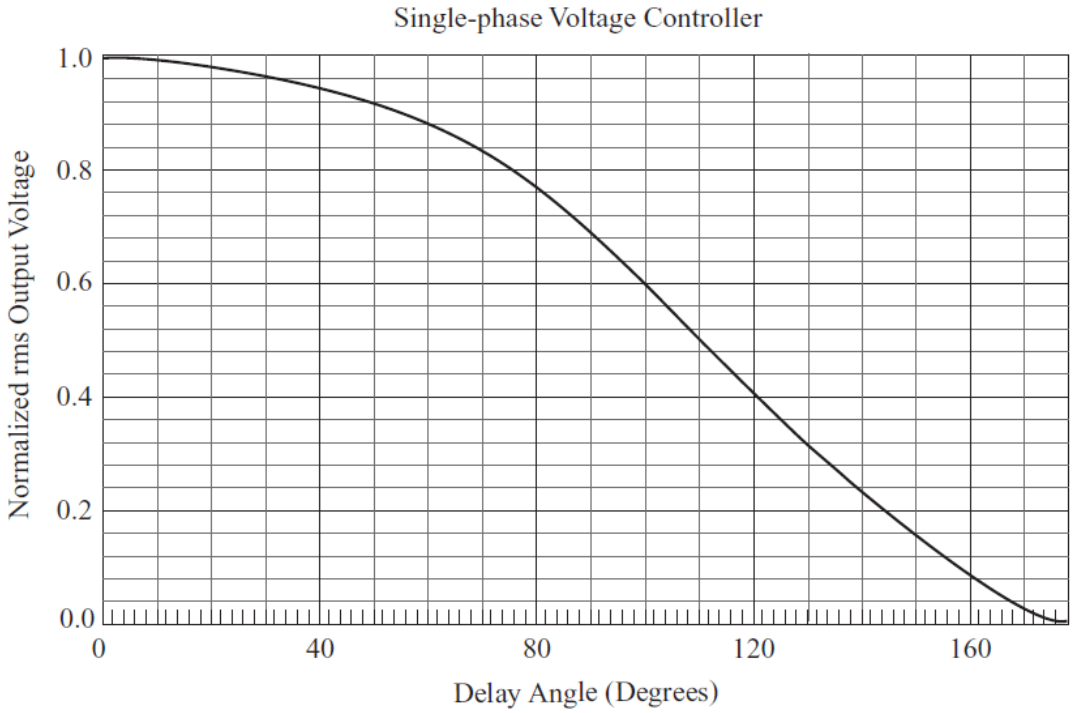


Figure 5-2 Normalized rms load voltage vs. delay angle for a single-phase ac voltage controller with a resistive load.

The rms current in the load and the source is:

$$I_{o,\text{rms}} = \frac{V_{o,\text{rms}}}{R} \quad (5\text{-}4)$$

And the power factor of the load is:

$$\text{pf} = \frac{P}{S} = \frac{P}{V_{s,\text{rms}} I_{s,\text{rms}}} = \frac{V_{o,\text{rms}}^2 / R}{V_{s,\text{rms}}(V_{o,\text{rms}}/R)} = \frac{V_{o,\text{rms}}}{V_{s,\text{rms}}}$$

$$= \frac{\frac{V_m}{\sqrt{2}}\sqrt{1 - \frac{\alpha}{\pi} + \frac{(\sin 2\alpha)}{2\pi}}}{V_m/\sqrt{2}}$$

$$\boxed{\text{pf} = \sqrt{1 - \frac{\alpha}{\pi} + \frac{\sin(2\alpha)}{2\pi}}} \qquad (5\text{-}5)$$

Note that $pf = 1$ for $\alpha = 0$, which is the same as for an uncontrolled resistive load, and the power factor for $\alpha > 1$is less than 1.
The average source current is zero because of half-wave symmetry. The average SCR current is:

$$I_{\text{SCR,avg}} = \frac{1}{2\pi}\int_{\alpha}^{\pi} \frac{V_m \sin(\omega t)}{R}\, d(\omega t) = \frac{V_m}{2\pi R}(1 + \cos\alpha) \qquad (5\text{-}6)$$

Since each SCR carries one-half of the line current, the rms current in each SCR is:

$$I_{\text{SCR,rms}} = \frac{I_{o,\text{rms}}}{\sqrt{2}} \qquad (5\text{-}7)$$

Since the source and load current is non-sinusoidal, harmonic distortion is a consideration when designing and applying AC voltage controllers. Only odd harmonics exist in the line current because the waveform has half-wave symmetry.
Harmonic currents are derived from the defining Fourier equations in Chap. 2.
Normalized harmonic content of the line currents versus α is shown in Fig. 5-3. Base current is source voltage divided by resistance, which is the current for $\alpha = 0$.

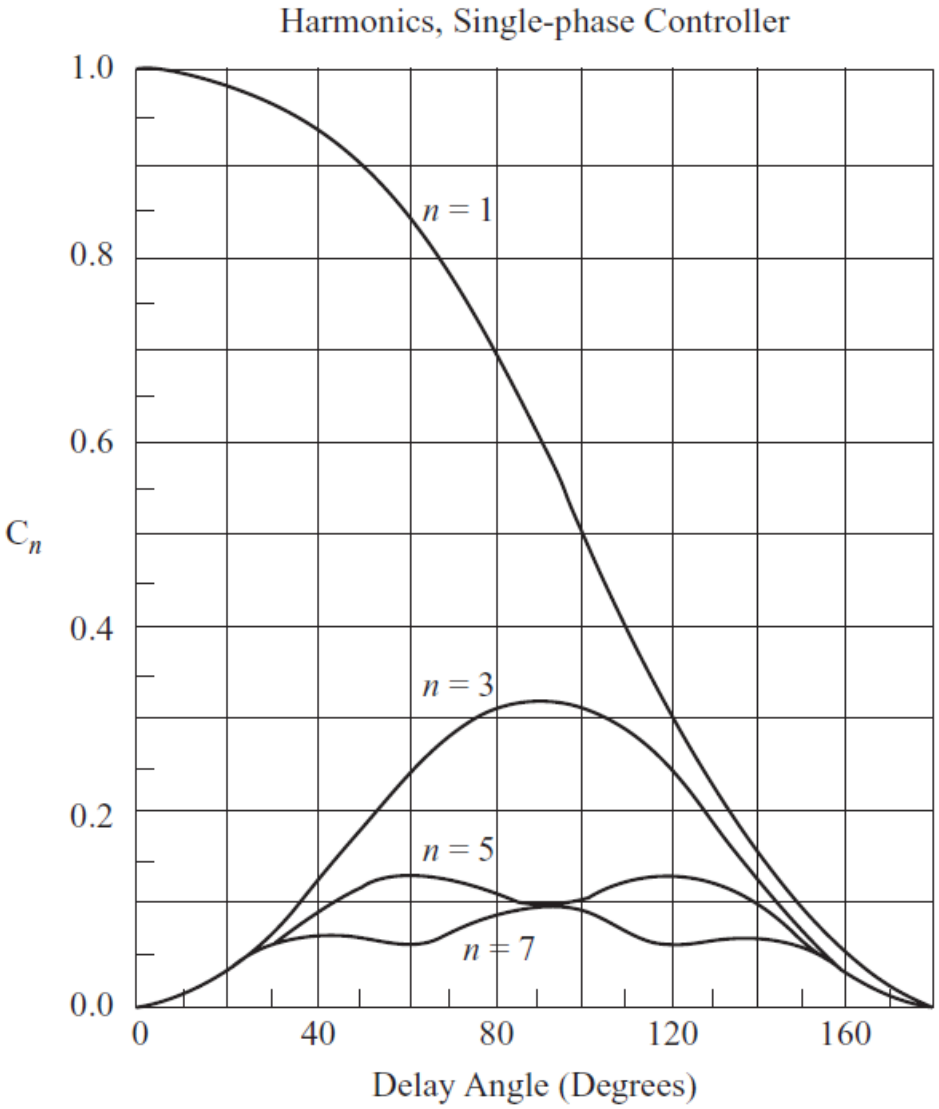


Figure 5-3 Normalized harmonic content versus delay angle for a single-phase AC voltage controller with a resistive load; C_n is the normalized amplitude. (See Chap. 2.)

Example 5-1 (Single-Phase half-wave AC voltage controller with a Resistive Load)

The single-phase AC voltage controller of Fig. 5-1a has a $120\ V\ rms\ \ 60\ Hz$ source. The load resistance is $15\ \Omega$. Determine (a) the delay angle required to deliver $500\ W$ to the load, (b) the rms source current, (c) the rms and average currents in the SCRs, (d) the power factor, and (e) the total harmonic distortion (THD) of the source current.

■ Solution

(a) The required rms voltage to deliver $500\ W$ to a $15\ \Omega$ load is:

$$P = \frac{V_{o,\text{rms}}^2}{R}$$

$$V_{o,\text{rms}} = \sqrt{PR} = \sqrt{(500)(15)} = 86.6 \text{ V}$$

The relationship between output voltage and delay angle is described by Eq. (5-3) and Fig. 5-2. From Fig. 5-2, the delay angle required to obtain a normalized output of $\frac{86.6}{120} = 0.72$ is approximately 90°. A more precise solution is obtained from the numerical solution for α in Eq. (5-3), expressed as:

$$86.6 - 120\sqrt{1 - \frac{\alpha}{\pi} + \frac{\sin(2\alpha)}{2\pi}} = 0$$

which yields

$$\alpha = 1.54 \text{ rad} = 88.1°$$

(b) Source rms current is

$$I_{o,\text{rms}} = \frac{V_{o,\text{rms}}}{R} = \frac{86.6}{15} = 5.77 \text{ A}$$

(c) SCR currents are determined from Eqs. (5-6) and (5-7),

$$I_{\text{SCR,rms}} = \frac{I_{\text{rms}}}{\sqrt{2}} = \frac{5.77}{\sqrt{2}} = 4.08 \text{ A}$$

$$I_{\text{SCR,avg}} = \frac{\sqrt{2}(120)}{2\pi(15)}\left[1 + \cos(88.1°)\right] = 1.86 \text{ A}$$

(d) The power factor is

$$\text{pf} = \frac{P}{S} = \frac{500}{(120)(5.77)} = 0.72$$

which could also be computed from Eq. (5-5).
(e) Base rms current is

$$I_{\text{base}} = \frac{V_{s,\text{rms}}}{R} = \frac{120}{15} = 8.0 \text{ A}$$

The rms value of the current's fundamental frequency is determined from C_1 in the graph of Fig. 5-3.

$$C_1 \approx 0.61 \Rightarrow I_{1,\text{rms}} = C_1 I_{\text{base}} = (0.61)(8.0) = 4.9 \text{ A}$$

The THD is computed from Eq. (2-68),

$$\text{THD} = \frac{\sqrt{I_{\text{rms}}^2 - I_{1,\text{rms}}^2}}{I_{1,\text{rms}}} = \frac{\sqrt{5.77^2 - 4.9^2}}{4.9} = 0.63 = 63\%$$

Single-Phase Controller with an *RL* Load
Figure 5-4*a* shows a single-phase ac voltage controller with an *RL* load.

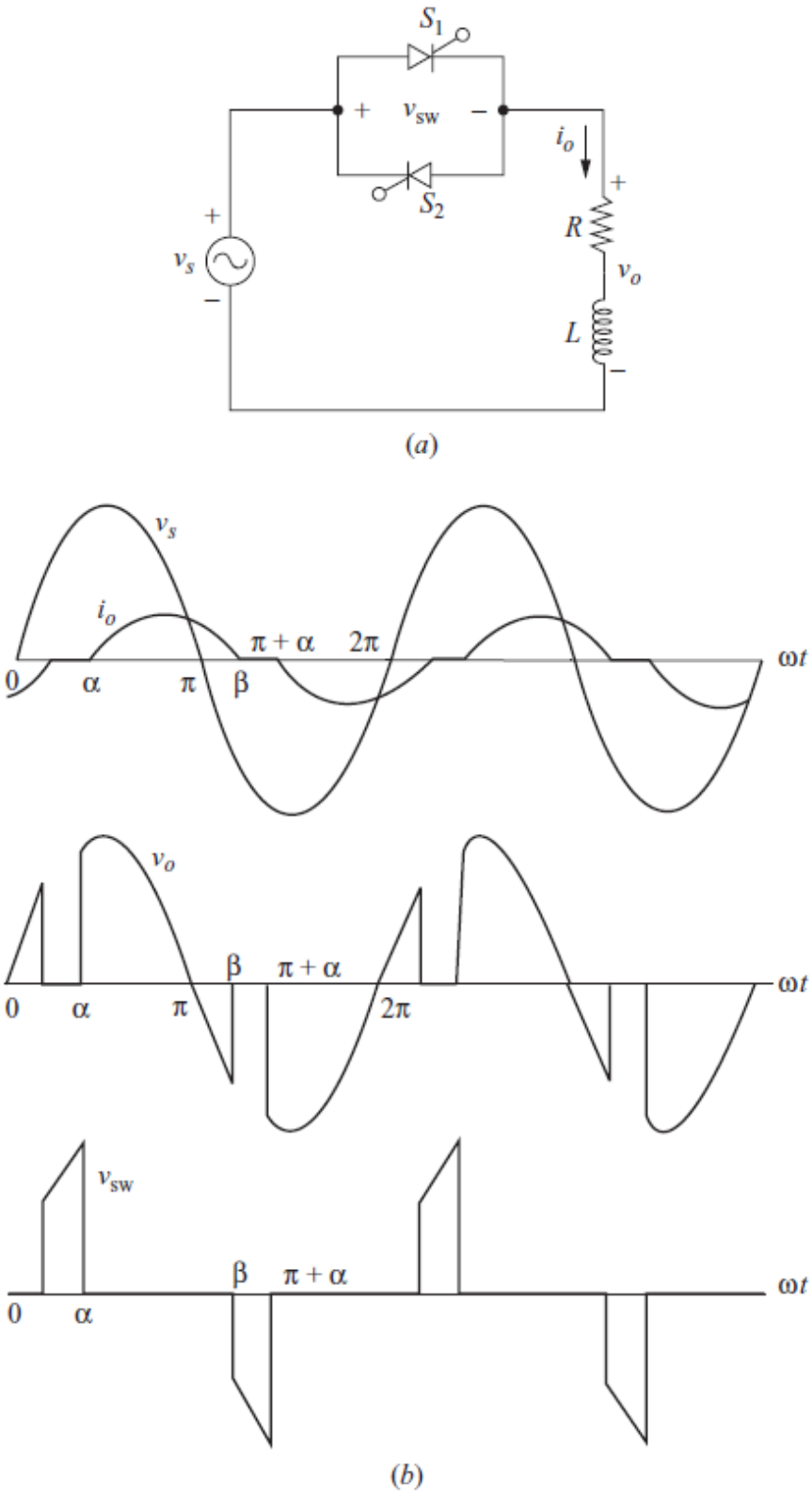


Figure 5-4 (*a*) Single-phase AC voltage controller with an *RL* load; (*b*) Typical waveforms.

When a gate signal is applied to S_1 at $\omega t = \alpha$, Kirchhoff's voltage law for the circuit is expressed as:

$$V_m \sin(\omega t) = R i_o(t) + L\frac{di_o(t)}{dt} \qquad (5\text{-}8)$$

The solution for current in this equation, outlined in Sec. 3.9, is:

$$i_o(\omega t) = \begin{cases} \frac{V_m}{Z}\left[\sin(\omega t - \theta) - \sin(\alpha - \theta)\, e^{(\alpha - \omega t)/\omega\tau}\right] & \text{for } \alpha \le \omega t \le \beta \\ 0 & \text{otherwise} \end{cases}$$

where

$$Z = \sqrt{R^2 + (\omega L)^2}, \quad \text{and} \quad \theta = \tan^{-1}\left(\frac{\omega L}{R}\right) \qquad (5\text{-}9)$$

The extinction angle β is the angle at which the current returns to zero, when $\omega t = \beta$,

$$i_o(\beta) = 0 = \frac{V_m}{Z}\left[\sin(\beta - \theta) - \sin(\alpha - \theta)\, e^{(\alpha - \beta)/\omega\tau}\right] \qquad (5\text{-}10)$$

which must be solved numerically for β.

Agate signal is applied to S_2 at $\omega t = \pi + \alpha$, and the load current is negative but has a form identical to that of the positive half-cycle. Figure 5-4b shows typical waveforms for a single-phase ac voltage controller with an RL load. The conduction angle γ is defined as:

$$\gamma = \beta - \alpha \qquad (5\text{-}11)$$

In the interval between π and β when the source voltage is negative and the load current is still positive, S_2 cannot be turned on because it is not forwardbiased.

The gate signal to S_2 must be delayed at least until the current in S_1 reaches zero, at $\omega t = \beta$. The delay angle is therefore at least $\beta - \pi$.

$$\alpha \ge \beta - \pi \qquad (5\text{-}12)$$

The limiting condition when $\beta - \alpha = \pi$is determined from an examination of Eq. (5-10). When $\alpha = \theta$, Eq. (5-10) becomes:

$$\sin(\beta - \alpha) = 0$$

which has a solution

$$\beta - \alpha = \pi$$

Therefore,

$$\gamma = \pi \qquad \text{when} \quad \alpha = \theta \tag{5-13}$$

If $\alpha < \theta,\ \gamma = \pi$, provided that the gate signal is maintained beyond $\omega t = \theta$. In the limit, when $\gamma = \pi$, one SCR is always conducting, and the voltage across the load is the same as the voltage of the source. The load voltage and current are sinusoids for this case, and the circuit is analyzed using phasor analysis for AC circuits. The power delivered to the load is continuously controllable between the two extremes corresponding to full source voltage and zero.

This SCR combination can act as a solid-state relay, connecting or disconnecting the load from the ac source by gate control of the SCRs. The load is disconnected from the source when no gate signal is applied, and the load has the same voltage as the source when a gate signal is continuously applied. In practice, the gate signal may be a high-frequency series of pulses rather than a continuous dc signal.

An expression for rms load current is determined by recognizing that the square of the current waveform repeats every π rad. Using the definition of rms,

$$I_{o,\text{rms}} = \sqrt{\frac{1}{\pi}\int_{\alpha}^{\beta} i_o^2(\omega t)\, d(\omega t)} \tag{5-14}$$

where $i_o(\omega t)$ is described in Eq. (5-9).

Power absorbed by the load is determined from

$$P = I_{o,\text{rms}}^2 R \tag{5-15}$$

The rms current in each SCR is:

$$I_{\text{SCR,rms}} = \frac{I_{o,\text{rms}}}{\sqrt{2}} \tag{5-16}$$

The average load current is zero, but each SCR carries one-half of the current waveform, making the average SCR current

$$I_{\text{SCR,avg}} = \frac{1}{2\pi}\int_{\alpha}^{\beta} i_o(\omega t)\, d(\omega t) \tag{5-17}$$

Example 5-2 (Single-Phase Voltage Controller with *RL* Load)

For the single-phase voltage controller of Fig. 5-4*a*, the source is $120\ V\ rms$ at 60 Hz, and the load is a series *RL* combination with $R = 20\ \Omega$ and $L =$

50 mH. The delay angle α is 90°. Determine (*a*) an expression for load current for the first half-period, (*b*) the rms load current, (*c*) the rms SCR current, (*d*) the average SCR current, (*e*) the power delivered to the load, and (*f*) the power factor.

■ **Solution**

(*a*) The current is expressed as in Eq. (5-9). From the parameters given,

$$Z = \sqrt{R^2 + (\omega L)^2} = \sqrt{(20)^2 + \left[(377)(0.05)\right]^2} = 27.5\,\Omega$$

$$\theta = \tan^{-1}\left(\frac{\omega L}{R}\right) = \tan^{-1}\frac{(377)(0.05)}{20} = 0.756 \text{ rad}$$

$$\omega\tau = \omega\left(\frac{L}{R}\right) = 377\left(\frac{0.05}{20}\right) = 0.943 \text{ rad}$$

$$\frac{V_m}{Z} = \frac{120\sqrt{2}}{27.5} = 6.18 \text{ A}$$

$$\alpha = 90° = 1.57 \text{ rad}$$

$$\frac{V_m}{Z}\sin(\alpha - \theta)\, e^{\alpha/\omega\tau} = 23.8 \text{ A}$$

The current is then expressed in Eq. (5-9) as

$$i_o(\omega t) = 6.18 \sin(\omega t - 0.756) - 23.8e^{-\omega t/0.943} \quad \text{A} \qquad \text{for } \alpha \le \omega t \le \beta$$

The extinction angle β is determined from the numerical solution of $i(\beta) = 0$ in the above equation, yielding:

$$\beta = 3.83 \text{ rad} = 220°$$

Note that the conduction angle $\gamma = \beta - \alpha = 2.26\,rad = 130°$, which is less than the limit of 180°.

(b) The rms load current is determined from Eq. (5-14).

$$I_{o,\text{rms}} = \sqrt{\frac{1}{\pi}\int_{1.57}^{3.83}\left[6.18 \sin(\omega t - 0.756) - 23.8e^{-\omega t/0.943}\right] d(\omega t)} = 2.71 \text{ A}$$

(c) The rms current in each SCR is determined from Eq. (5-16).

$$I_{\text{SCR,rms}} = \frac{I_{o,\text{rms}}}{\sqrt{2}} = \frac{2.71}{\sqrt{2}} = 1.92 \text{ A}$$

(d) Average SCR current is obtained from Eq. (5-17).

$$I_{\text{SCR, avg}} = \frac{1}{2\pi}\int_{1.57}^{3.83}\left[6.18 \sin(\omega t - 0.756) - 23.8e^{-\omega t/0.943}\right] d(\omega t) = 1.04 \text{ A}$$

(e) Power absorbed by the load is

$$P = I_{o,\text{rms}}^2 R = (2.71)^2(20) = 147 \text{ W}$$

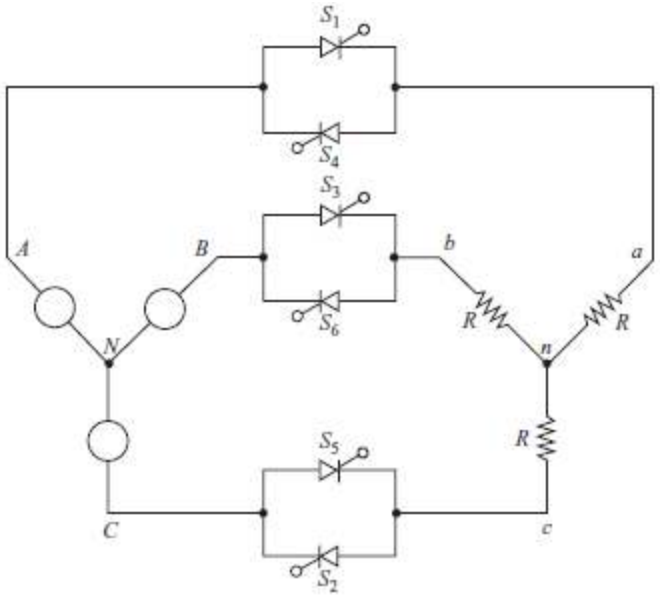
(f) Power factor is determined from P/S.

$$\text{pf} = \frac{P}{S} = \frac{P}{V_{s,\text{rms}} I_{s,\text{rms}}} = \frac{147}{(120)(2.71)} = 0.45 = 45\%$$

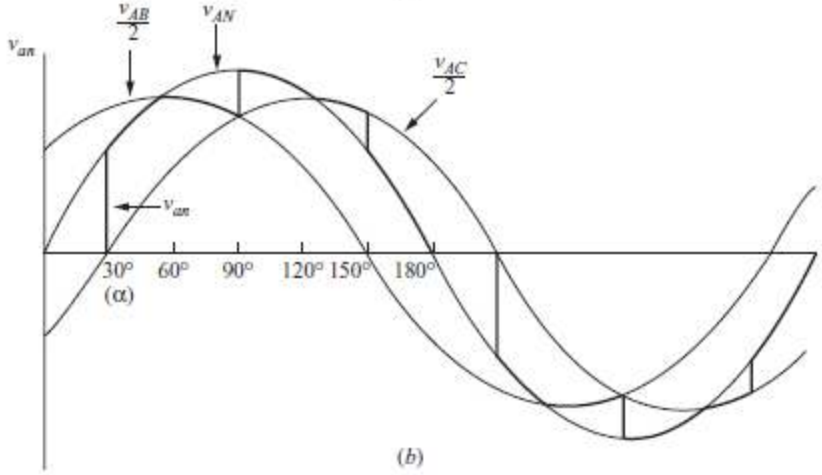
5.3 Three-Phase Voltage Controllers

Y-Connected Resistive Load

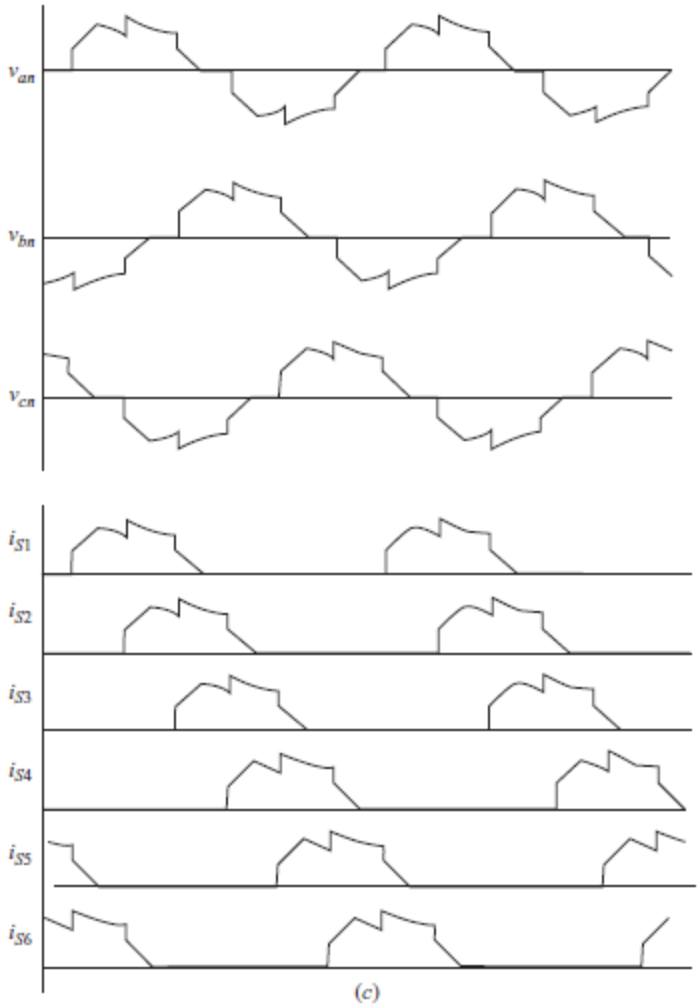
A three-phase voltage controller with a *Y*-connected resistive load is shown in Fig. 5-7*a*.


S_1
S_4
S_3
S_6
S_5
S_2
A
B
N
C
b
a
n
c
R
R
R

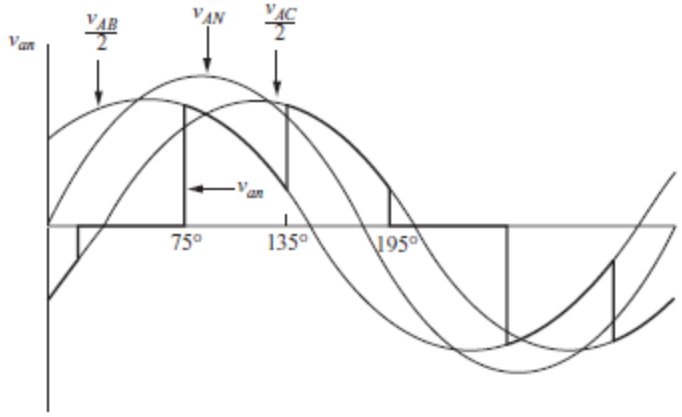
(a)


v_{an}
$\frac{v_{AB}}{2}$
v_{AN}
$\frac{v_{AC}}{2}$
v_{an}
30°
(α)
60°
90°
120°
150°
180°

(b)


v_{an}
v_{bn}
v_{cn}
i_{S1}
i_{S2}
i_{S3}
i_{S4}
i_{S5}
i_{S6}

(c)


v_{an}
$\frac{v_{AB}}{2}$
v_{AN}
$\frac{v_{AC}}{2}$
v_{an}
75°
135°
195°

(d)

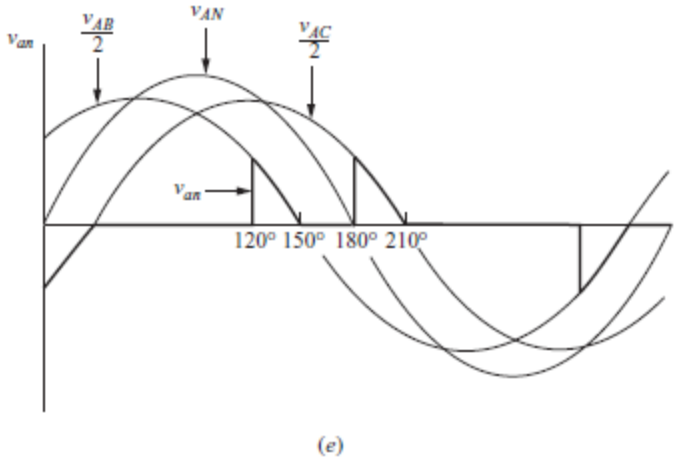


Figure 5-7 (*a*) Three-phase ac voltage controller with a *Y*-connected resistive load; (*b*) Load voltage $v_{an}$for $\alpha = 30°$; (*c*) Load voltages and switch currents for a three-phase resistive load for $\alpha = 30°$; (*d*) Load voltage v_{an} for $\alpha = 75°$; (*e*) Load voltage v_{an} for $\alpha = 120°$

The power delivered to the load is controlled by the delay angle α on each SCR. The six SCRs are turned on in the sequence 1-2-3-4-5-6, at $60°$ intervals.
Gate signals are maintained throughout the possible conduction angle.
The instantaneous voltage across each phase of the load is determined by which SCRs are conducting. At any instant, three SCRs, two SCRs, or no SCRs are on. The instantaneous load voltages are either a line-to-neutral voltage (three on), one-half of a line-to-line voltage (two on), or zero (none on).
When three SCRs are on (one in each phase), all three phase voltages are connected to the source, corresponding to a balanced three-phase source connected to a balanced three-phase load. The voltage across each phase of the load is the corresponding line-to-neutral voltage. For example, if S_1, S_2, and S_6 are on, $v_{an} = v_{AN}$, $v_{bn} = v_{BN}$, and $v_{cn} = v_{CN}$. When two SCRs are on, the line-to-line voltage of those two phases is equally divided between the two load resistors that are connected.
For example, if only S_1 and S_2 are on, $v_{an} = v_{AC}/2$, $v_{cn} = v_{CA}/2$, and $v_{bn} = 0$.
Which SCRs are conducting depends on the delay angle _ and on the source voltages at a particular instant. The following are the ranges of α that produce particular types of load voltages with an example for each:
For $0 < \alpha < 60°$:

Two or three SCRs conduct at any one time for this range of α. Figure 5-7b shows the load line-to-neutral voltage v_{an} for $\alpha = 30°$. At $\omega t = 0$, S_5 and S_6 are conducting and there is no current in R_a, making $v_{an} = 0$. At ωt =π/6 (30°), S_1 receives a gate signal and begins to conduct; S_5 and S_6 remain on, and $v_{an} = v_{AN}$. The current in S_5 reaches zero at 60°, turning S_5 off. With S_1 and S_6 remaining on, $v_{an} = v_{AB}/2$. At 90°, S_2 is turned on; the three SCRs S_1, S_2, and S_6 are then on; and $v_{an} = v_{AN}$. At 120°, S_6 turns off, leaving S_1 and S_2 on, so $v_{an} = v_{AC}/2$. As the firing sequence for the SCRs proceeds, the number of SCRs on at a particular instant alternates between 2 and 3. All three phase-to neutral load voltages and switch currents are shown in Fig. 5-7c. For intervals to exist when three SCRs are on, the delay angle must be less than 60°.

$For\ 60° < \alpha < 90°$:

Only two SCRs conduct at any one time when the delay angle is between 60° and 90°. $Load = voltage\ v_{an}\ for\ \alpha = 75°$ is shown in Fig. 5-7*d*. Just prior to 75°, S_5 and S_6 are conducting, and $v_{an} = 0$. When S_1 is turned on at 75°, S_6 continues to conduct, but S_5 must turn off because v_{CN} is negative.

Voltage v_{an} is then $v_{AB}/2$. When S_2 is turned on at 135°, S_6 is forced off, and $v_{an} = v_{AC}/2$. The next SCR to turn on is S_3, which forces S_1 off, and $v_{an} = 0$. One SCR is always forced off when an SCR is turned on forα in this range. Load voltages are one-half line-to-line voltages or zero.

For $90° < \alpha < 150°$:

Only two SCRs can conduct at any one time in this mode. Additionally, there are intervals when no SCRs conduct. Figure 5-7*e* shows the load voltage v_{an} for $\alpha = 120°$. In the interval just prior to 120°, no SCRs are on, and $v_{an} = 0$.

At $\alpha = 120°$, S_1 is given a gate signal, and S_6 still has a gate signal applied. Since v_{AB} is positive, both S_1 and S_6 are forward-biased and begin to conduct, and $v_{an} = v_{AB}/2$. Both $S_1$and S_6 turn off when v_{AB} becomes negative. When a gate signal is applied to S_2, it turns on, and S_1 turns on again.

For $\alpha > 150°$, there is no time interval when an SCR is forward-biased while a gate signal is applied. Output voltage is zero for this condition.

Normalized output voltage viruses delay angle is shown in Fig. 5-8.

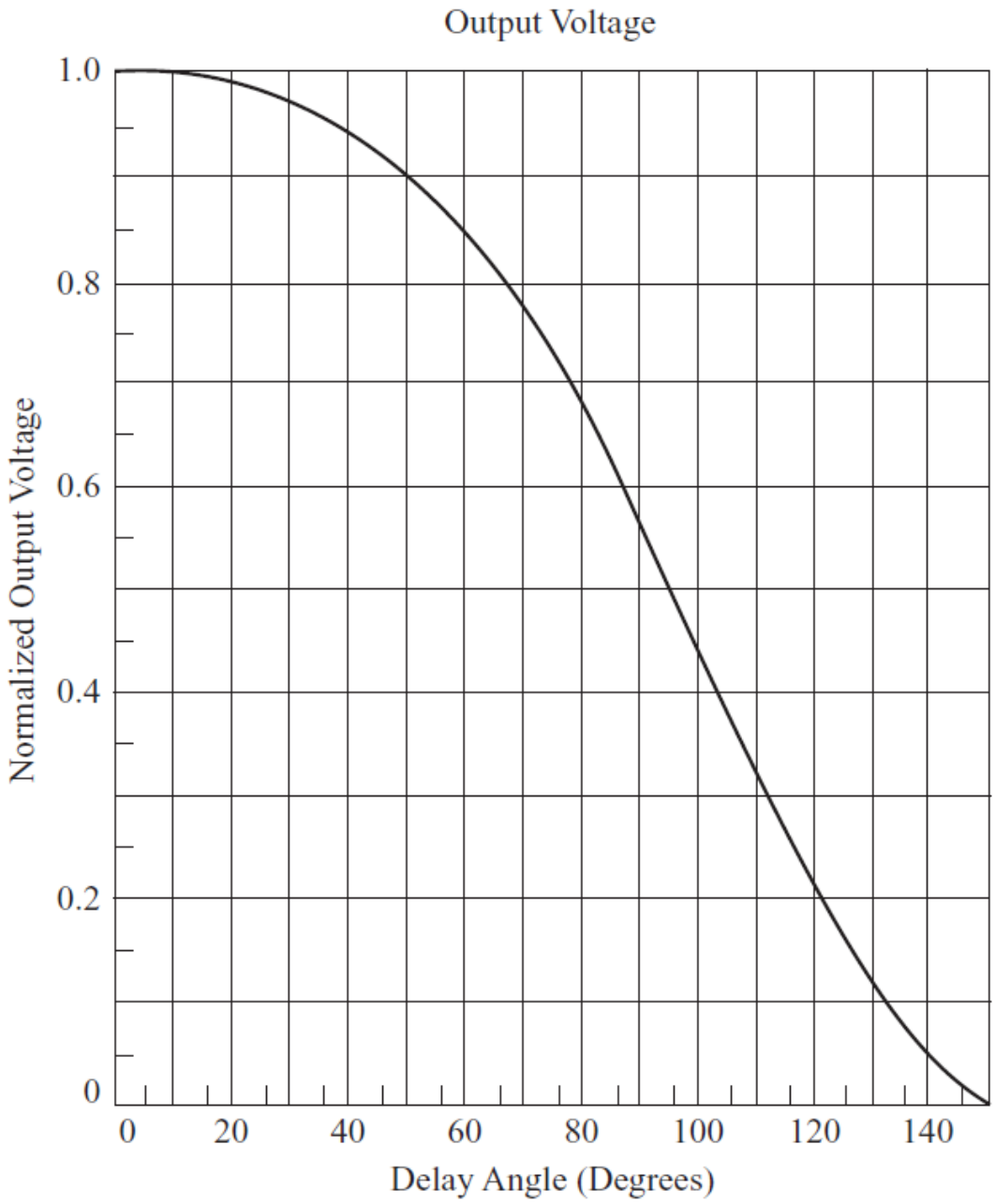


Figure 5-8 Normalized rms output voltage for a three-phase AC voltage controller with a resistive load.

Note that a delay angle of zero corresponds to the load being connected directly to the three-phase source. The range of output voltage for the three-phase voltage controller is between full source voltage and zero.
Harmonic currents in the load and line for the three-phase ac voltage controller are the odd harmonics of order $6n \pm 1, n = 1, 2, 3, \ldots$ (that is, 5th, 7th, 11th, 13th). . . . Harmonic filters may be required in some

applications to prevent harmonic currents from propagating into the ac system.
Since analysis of the three-phase ac voltage controller is cumbersome, simulation is a practical means of obtaining rms output voltages and power delivered to a load. PSpice simulation is presented in Example 5-4.

Y-Connected *RL* Load

The load voltages for a three-phase voltage controller with an *RL* load are again characterized by being a line-to-neutral voltage, one-half of a line-to-line voltage, or zero. The analysis is much more difficult for an *RL* load than for a resistive load, and simulation provides results that would be extremely difficult to obtain analytically.

Delta-Connected Resistive Load

Athree-phase ac voltage controller with a delta-connected resistive load is shown in Fig. 5-10*a*.

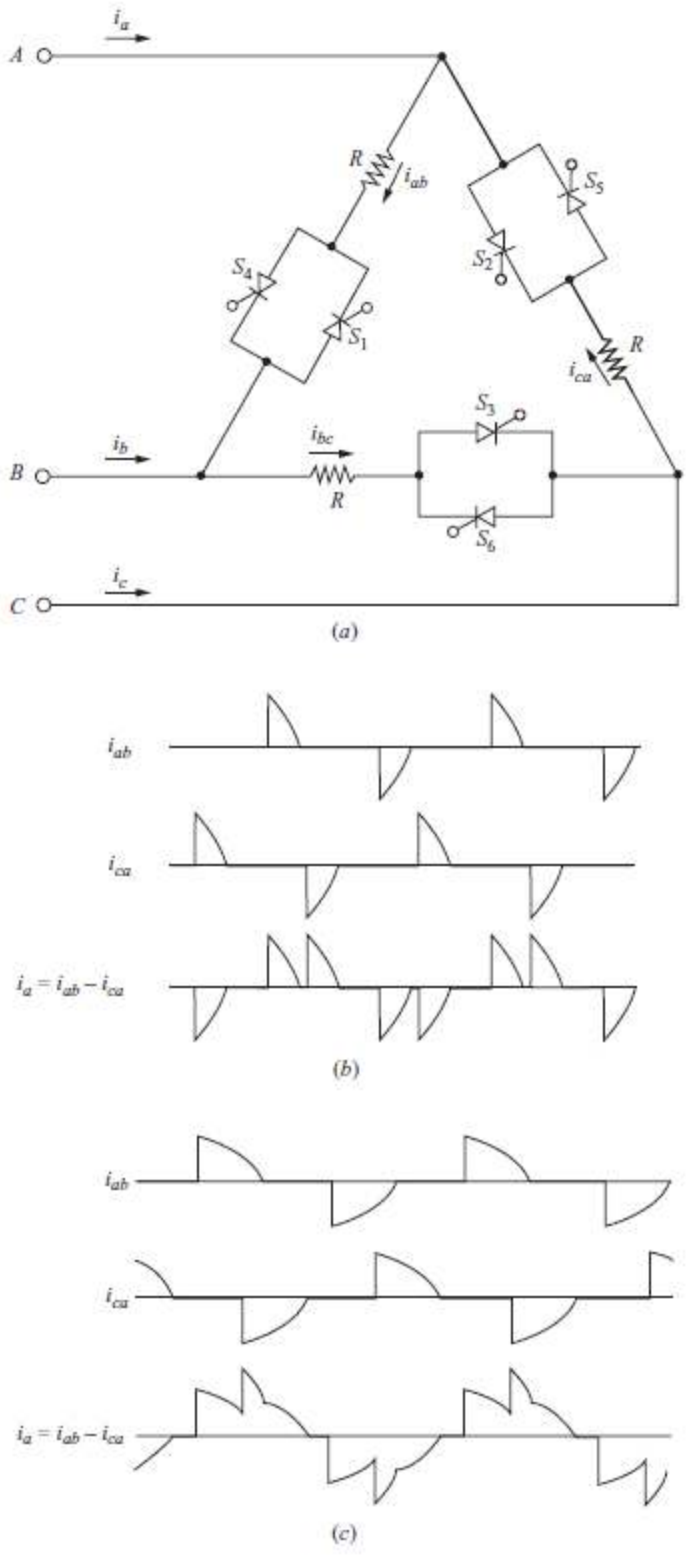


Figure 5-10 (*a*) Three-phase AC voltage controller with a delta-connected resistive load; (*b*) Current waveforms for $\alpha = 130°$; (*c*) Current waveforms for $\alpha = 90°$.

The voltage across a load resistor is the corresponding line-to-line voltage when a SCR in the phase is on. The delay angle is referenced to the zero crossing of the line-to-line voltage. SCRs are turned on in the sequence 1-2-3-4-5-6.
The line current in each phase is the sum of two of the delta currents:

$$\begin{aligned} i_a &= i_{ab} - i_{ca} \\ i_b &= i_{bc} - i_{ab} \\ i_c &= i_{ca} - i_{bc} \end{aligned} \qquad (5\text{-}18)$$

The relationship between rms line and delta currents depends on the conduction angle of the SCRs. For small conduction angles (large α), the delta currents do not overlap (Fig. 5-10b), and the rms line currents are:

$$I_{L,\text{rms}} = \sqrt{2}\, I_{\Delta,\text{rms}} \qquad (5\text{-}19)$$

For large conduction angles (small α), the delta currents overlap (Fig. 5-10c), and the rms line current is larger than $\sqrt{2}\ I_{\Delta}$. In the limit when $\gamma = \pi(\alpha = 0)$, the delta currents and line currents are sinusoids. The rms line current is determined from ordinary three-phase analysis.

$$I_{L,\text{rms}} = \sqrt{3}\, I_{\Delta,\text{rms}} \qquad (5\text{-}20)$$

The range of rms line current is therefore

$$\sqrt{2}\ I_{\Delta,\text{rms}} \leq I_{L,\text{rms}} \leq \sqrt{3}\ I_{\Delta,\text{rms}} \qquad (5\text{-}21)$$

Depending on α.
Use of the delta-connected three-phase voltage controller requires the load to be broken to allow thyristors to be inserted in each phase, which is often not feasible.

5.4 Induction Motor Speed Control

Squirrel-cage induction motor speed can be controlled by varying the voltage and/or frequency. The AC voltage controller is suitable for some speed-control applications. The torque produced by an induction motor is proportional to the square of the applied voltage. Typical torque-speed curves for an induction motor are shown in Fig. 5-11. If a load has a torque-speed characteristic like that also shown in Fig. 5-11, speed can be controlled by adjusting the motor voltage.

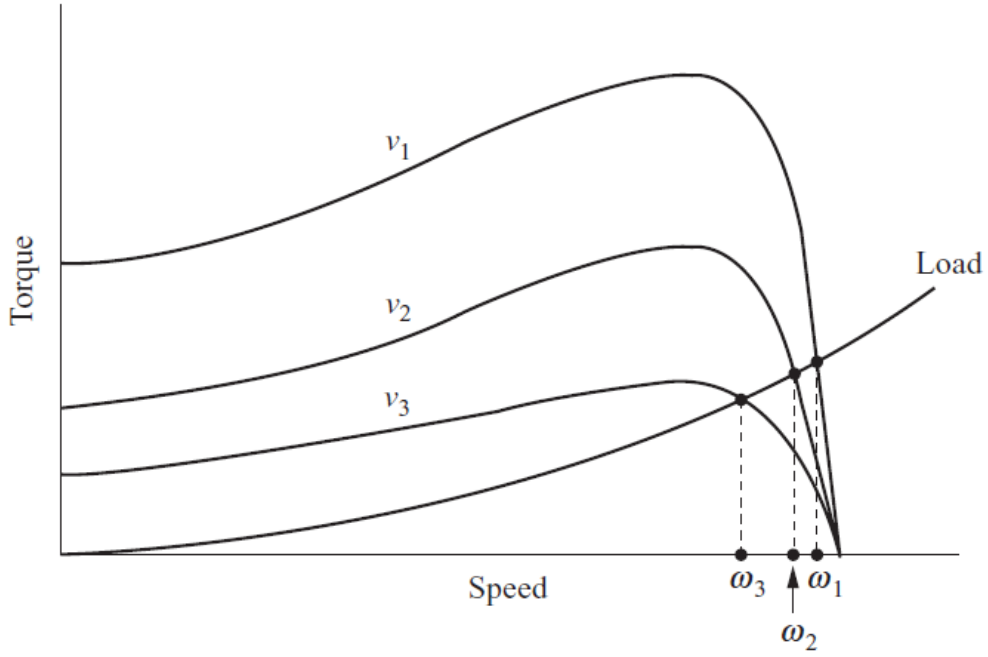


Figure 5-11 Torque-speed curves for an induction motor.

Operating speed corresponds to the intersection of the torque-speed curves of the motor and the load. A fan or pump is a suitable load for this type of speed control, where the torque requirement is approximately proportional to the square of the speed.

Single-phase induction motors are controlled with the circuit of Fig. 5-4*a*, and three-phase motors are controlled with the circuit of Fig. 5-7*a*. Energy efficiency is poor when using this type of control, especially at low speeds. The large slip at low speeds results in large rotor losses. Typical applications exist where the load is small, such as single-phase fractional-horsepower motors, or where the time of low-speed operation is short. Motor speed control using a variable-frequency source from an inverter circuit (Chap. 8) is usually a preferred method.

5.5 Static VAR Control

Capacitors are routinely placed in parallel with inductive loads for power factor improvement. If a load has a constant reactive voltampere (VAR) requirement, a fixed capacitor can be selected to correct the power factor to unity. However, if a load has a varying VAR requirement, the fixed-capacitor arrangement results in a changing power factor.

The circuit of Fig. 5-12 represents an application of an AC voltage controller to maintain a unity power factor for varying load VAR requirements.

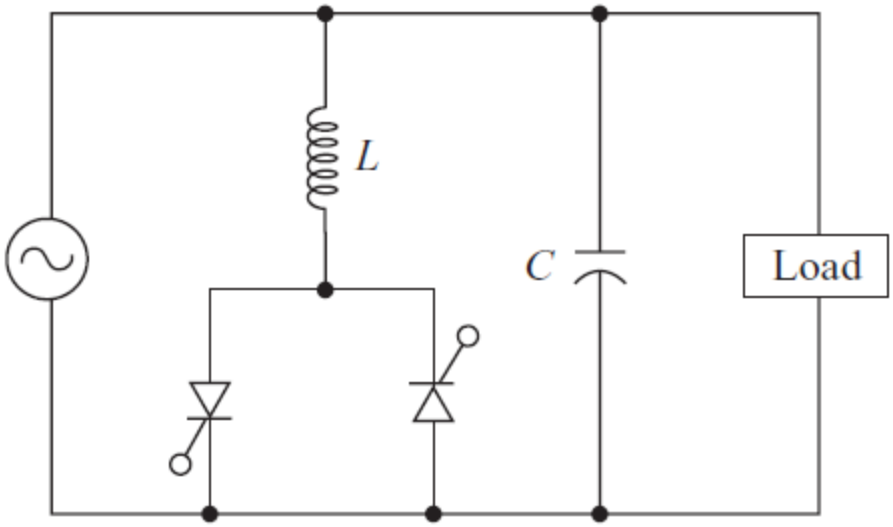


Figure 5-12 Static VAR control.

The power factor correction capacitance supplies a fixed amount of reactive power, generally greater than required by the load. The parallel inductance absorbs a variable amount of reactive power, depending on the delay angle of the SCRs. The net reactive power supplied by the inductor-capacitor combination is controlled to match that absorbed by the load. As the VAR requirement of the load changes, the delay angle is adjusted to maintain unity power factor. This type of power factor correction is known as *static VAR control*.

The SCRs are placed in the inductor branch rather than in the capacitor branch because very high currents could result from switching a capacitor with a SCR.

Static VAR control has the advantage of being able to adjust to changing load requirements very quickly. Reactive power is continuously adjustable with static VAR control, rather than having discrete levels as with capacitor banks which are switched in and out with circuit breakers. Static VAR control is becoming increasingly prevalent in installations with rapidly varying reactive power requirements, such as electric arc furnaces. Filters are generally required to remove the harmonic currents generated by the switched inductance.

5.6 Summary

• Voltage controllers use electronic switches to connect and disconnect a load to an ac source at regular intervals. This type of circuit is classified as an ac-ac converter.

• Voltage controllers are used in applications such as single-phase light-dimmer circuits, single-phase or three-phase induction motor control, and static VAR control.
• The delay angle for the thyristors controls the time interval for the switch being on and thereby controls the effective value of voltage at the load. The range of control for load voltage is between full ac source voltage and zero.
• An ac voltage controller can be designed to function in either the fully on or fully off mode. This application is used as a solid-state relay.
• The load and source current and voltage in ac voltage controller circuits may contain significant harmonics. For equal delay angles in the positive and negative half-cycles, the average source current is zero, and only odd harmonics exist.

- Three-phase voltage controllers can have Y- or Δ-connected loads.

• Simulation of single-phase or three-phase voltage controllers provides an efficient analysis method.